

Ice Property Measurement Using a Camera System in the IceCube Upgrade

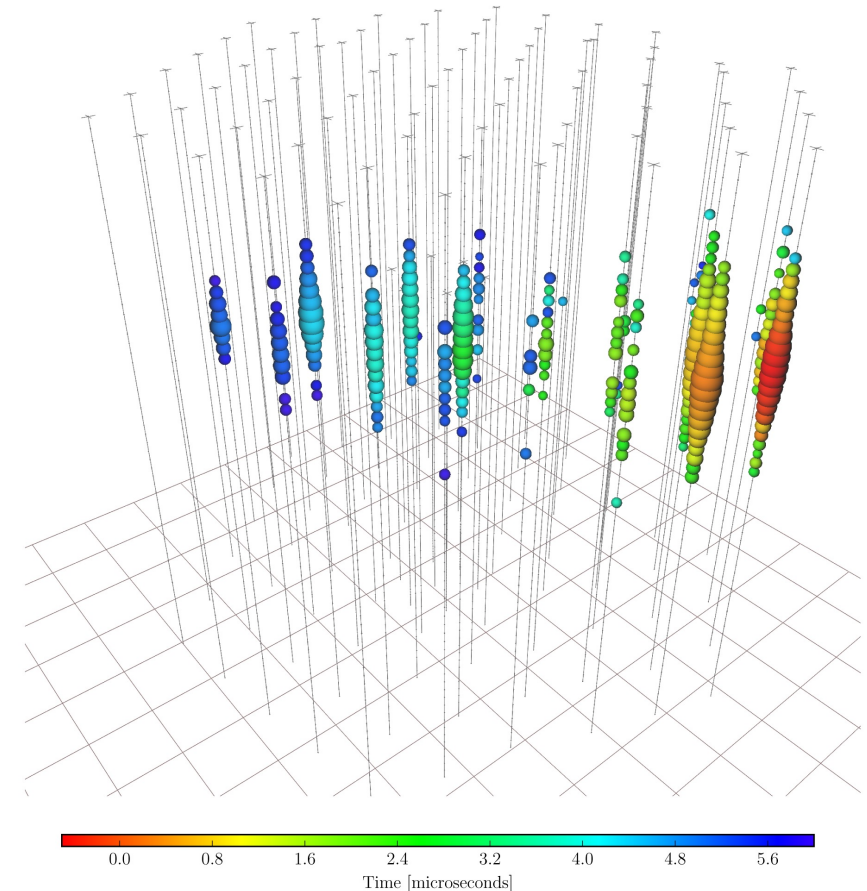
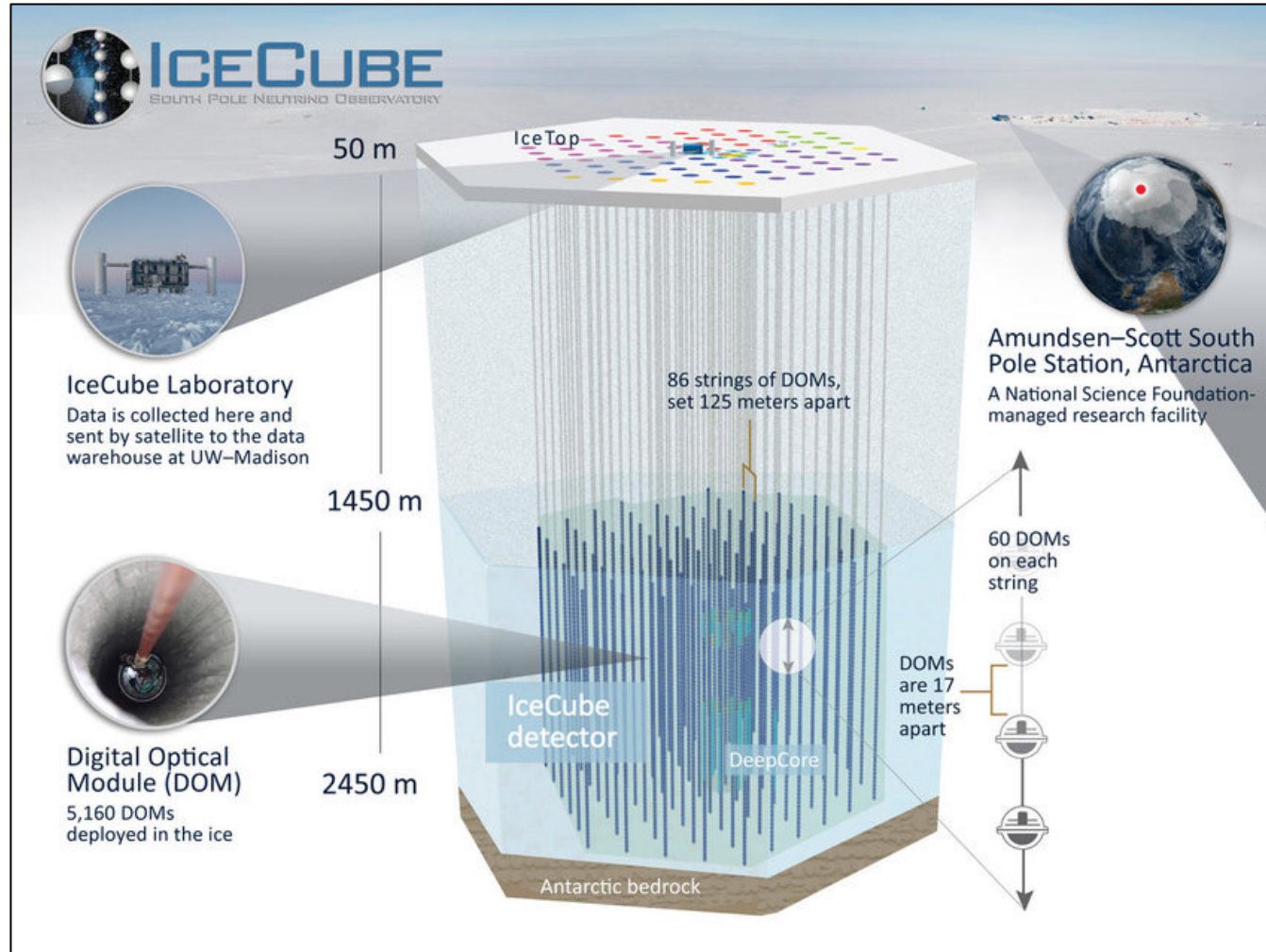
Minje Park

Carsten Rott, Steven Rodan, Matti Jansson ,
Woosik Kang, Minyeong Seo, Yubin Oh,
Seowon Choi, Yoonyoung Kim, Youngmin Kim

IceCube Neutrino Observatory

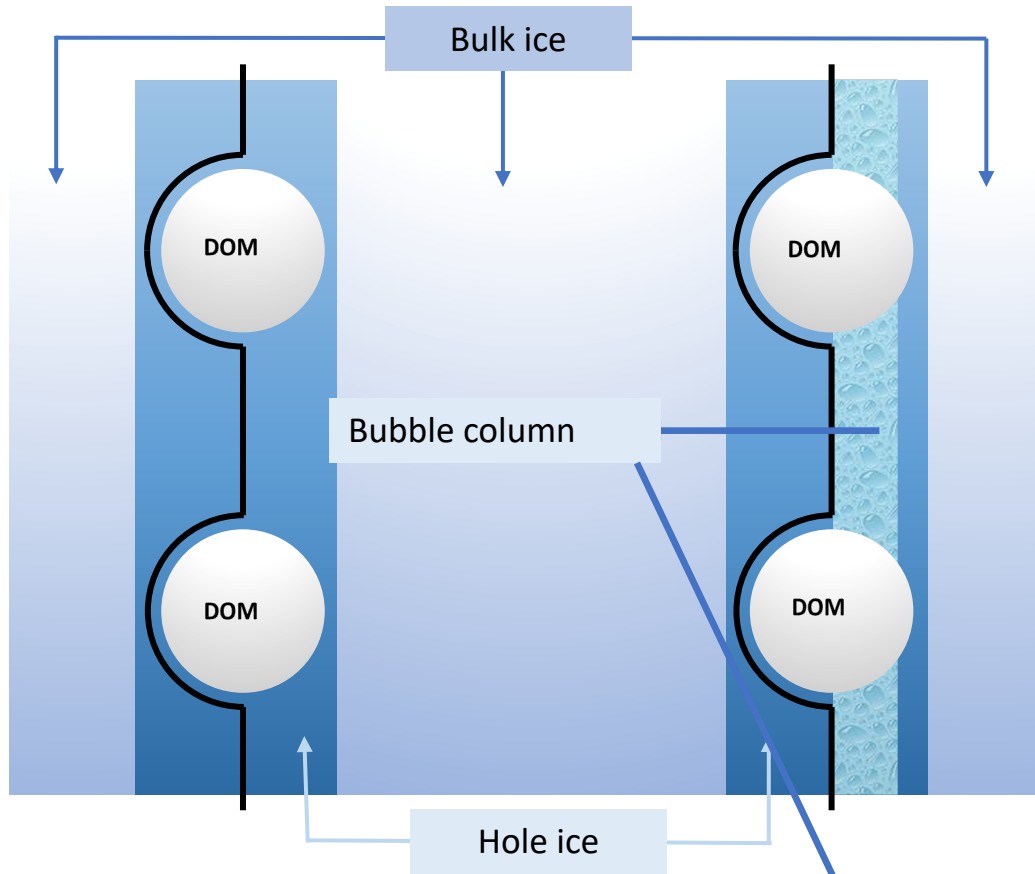
2

- Detecting astrophysical neutrino fluxes and identifying their sources.
- When neutrinos interact inside or near the detector volume, they produce charged secondary particles that emit Cherenkov photons, some of which are collected by the Digital Optical Modules (DOMs).



Optical Properties of Antarctic Ice

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Scattering Length :

- The average distance a photon travels before being scattered.

Absorption Length :

- The average distance a photon travels before being absorbed.

Main Bulk Ice Optical Properties

Main Bulk Ice Optical Properties	
Scattering Length	Absorption Length

Main Hole Ice Optical Properties

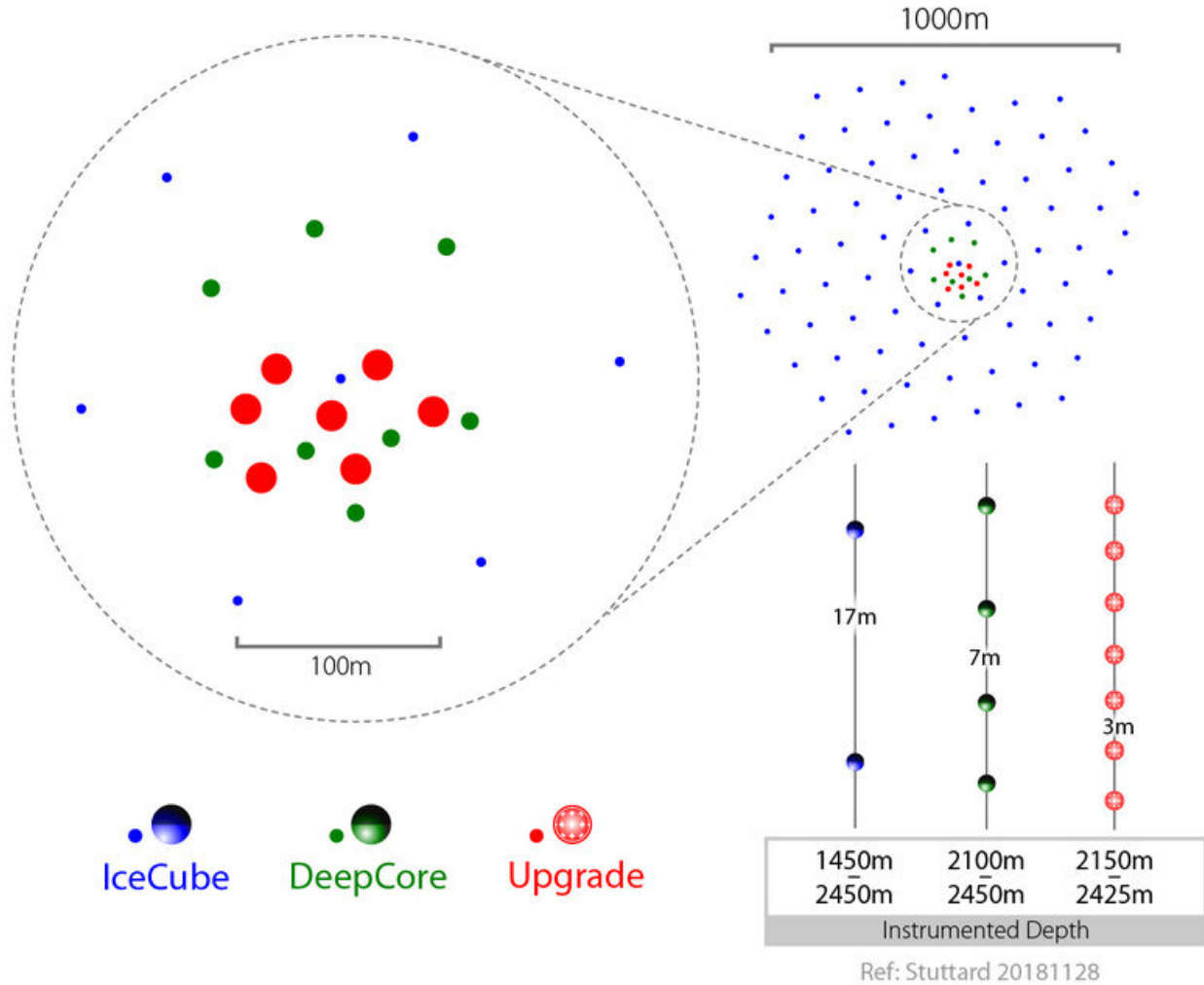
Scattering Length	Bubble column	
	Size & position	Scattering length

Bulk Ice:

- The pure Antarctic glacial ice

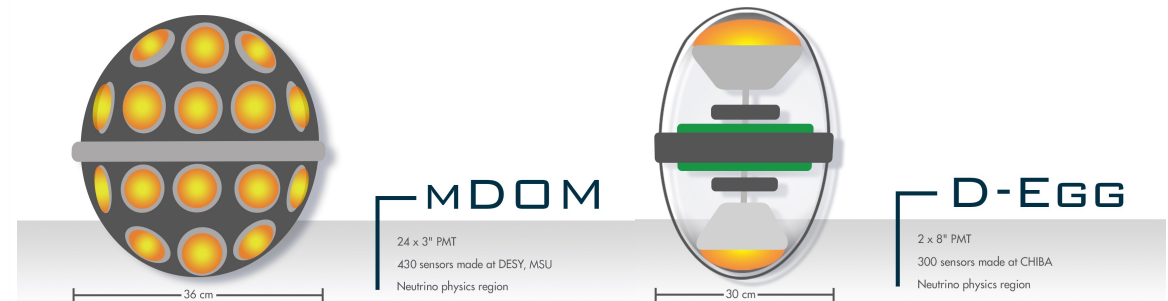
Hole Ice:

- The refrozen ice in which the strings are embedded. It contains residual air bubbles, increasing optical scattering around the DOMs.



Objectives

- Improve Low energy(1~100GeV) event resolution
- Improve our Measure the optical properties of the detector medium .



Details of the upgrade

- 7 additional cables in the core
- 700 new designed Doms
- Additional calibration systems (Camera System, etc)
- Construction : 2025~26

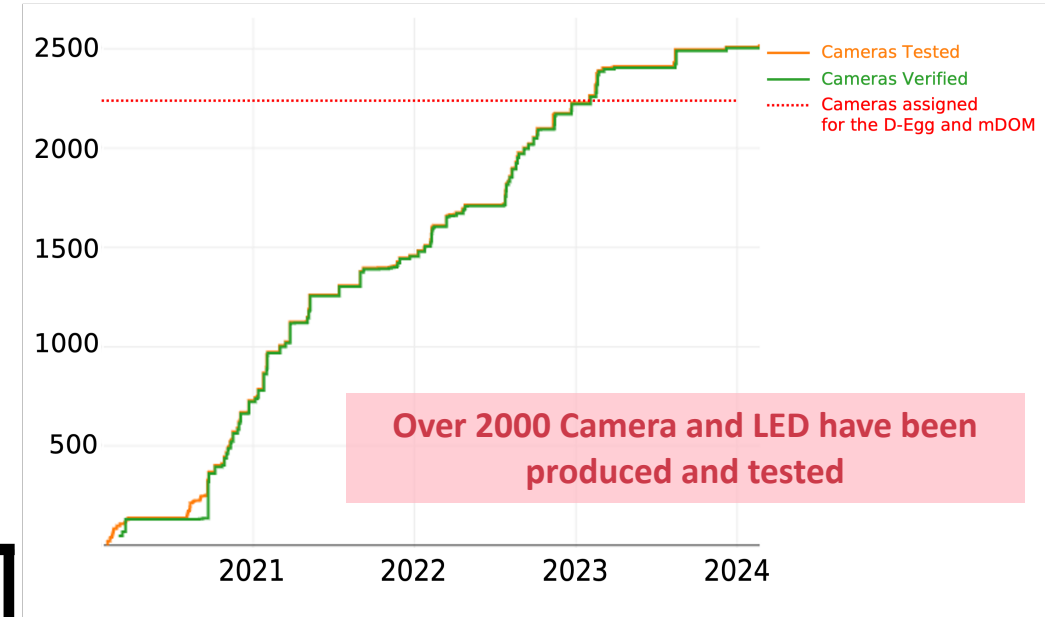
IceCube Upgrade Camera System

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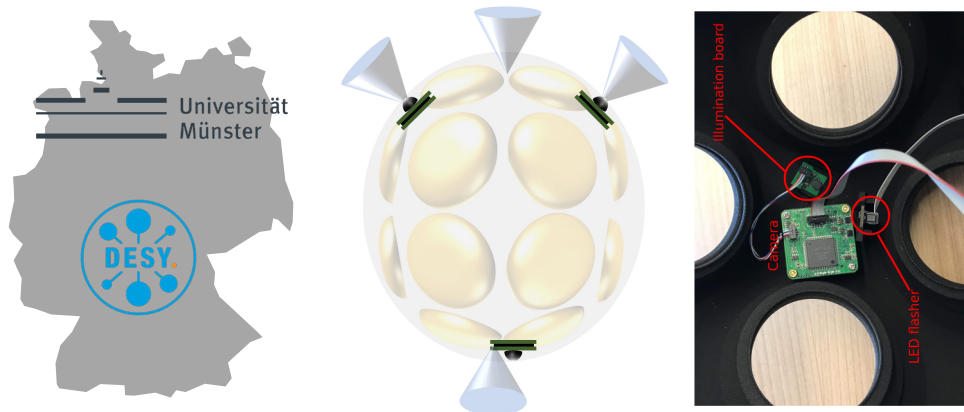


Developed and
Produced by **SKKU**

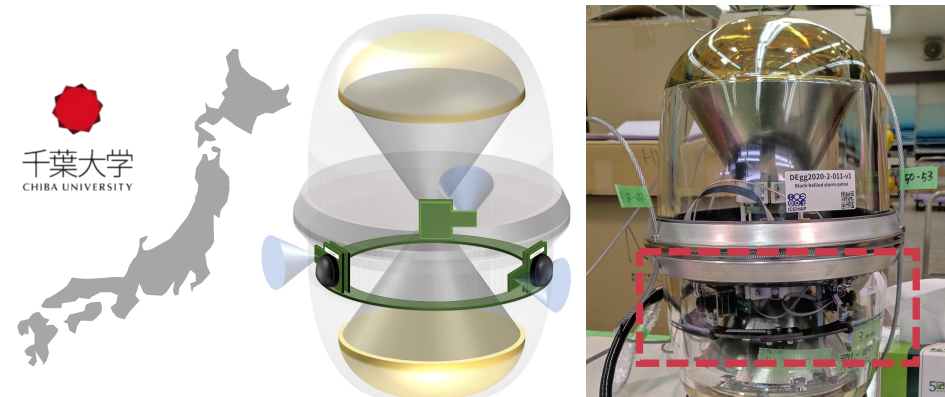
Acceptance Test



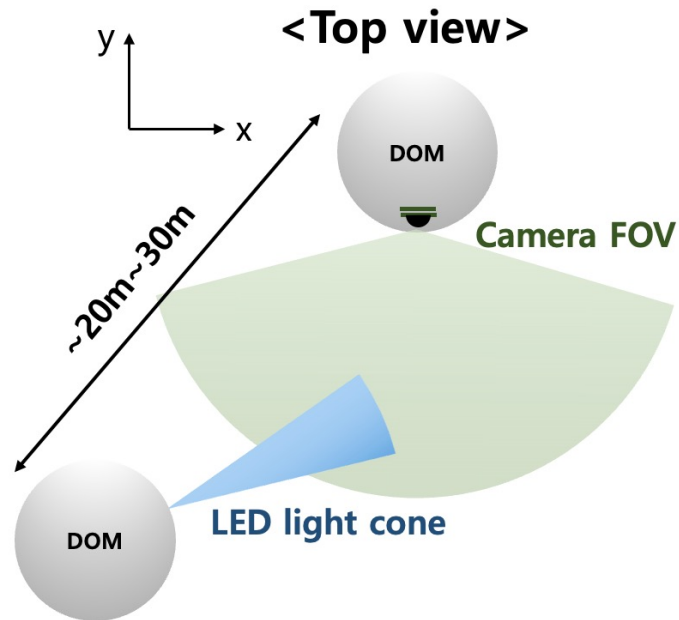
Sent to be Integrated into every **277** mDoms



Sent to be Integrated into every **402** D- Eggs



Bulk Ice Measurement

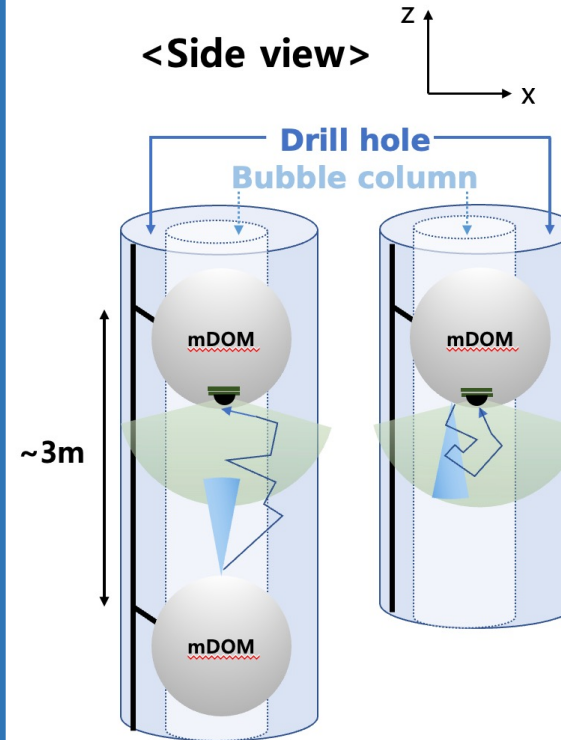


Bulk Ice

Scattering Length

Absorption Length

Hole Ice Measurement



- Capturing light from the LED on the DOM below
- Capturing light from the LED right next to the camera

Hole Ice

Scattering Length

Bubble column

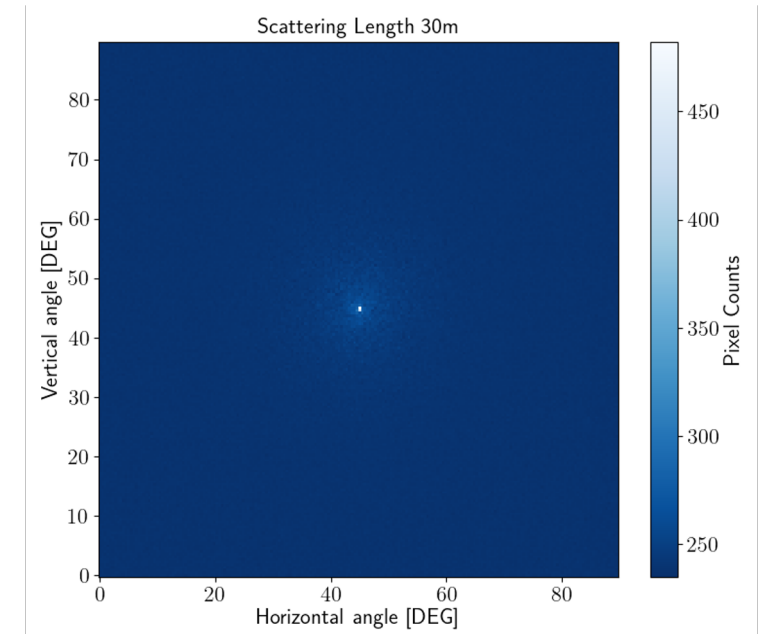
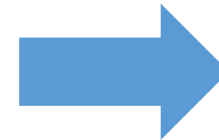
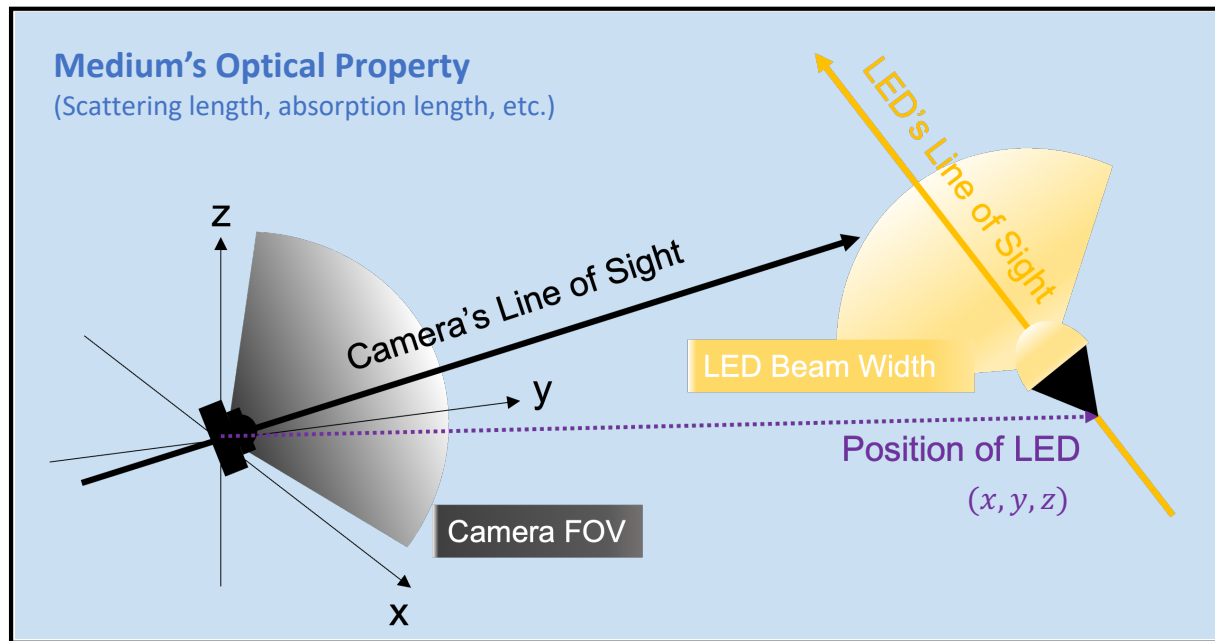
Size & Position

Scattering Length

Camera Simulation

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- CamSim is a camera image simulation framework based on Monte Carlo Photon Propagation Script.
- Configured with the positions and orientation of the camera and LED, and the optical properties of the ice.
- Simulates the number and positions of photons reaching the camera to produce the simulated images.



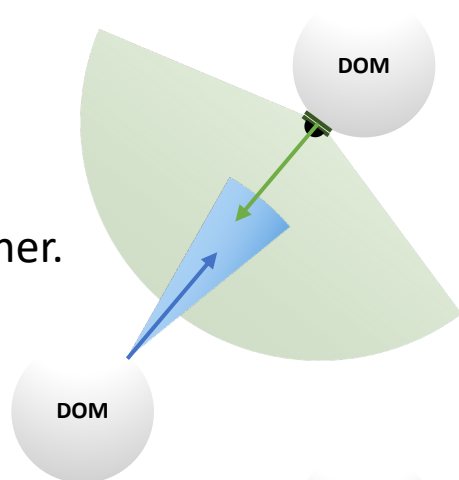
Measuring Bulk Ice Scattering Length

Using CNN

Simulate on two different geometric configurations with varied bulk ice scattering lengths.

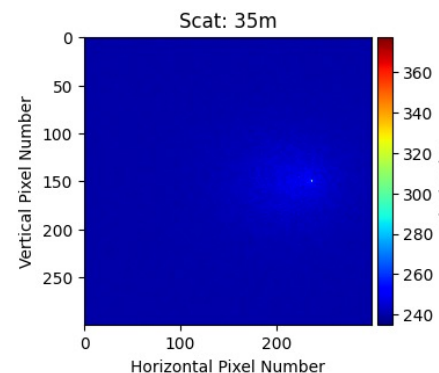
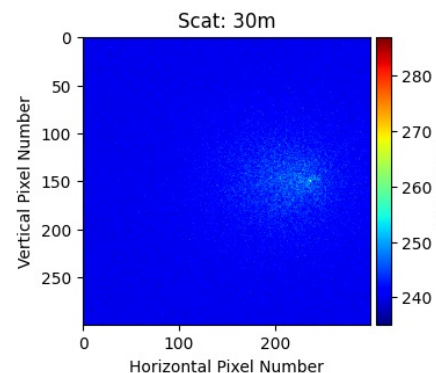
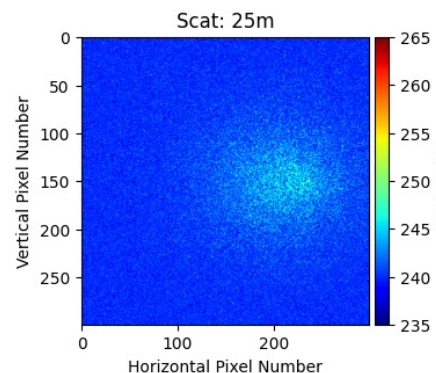
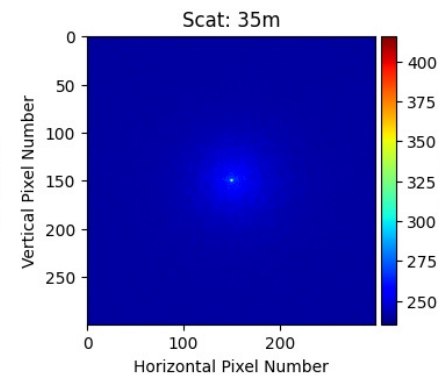
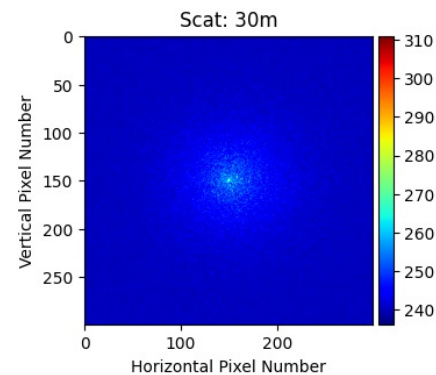
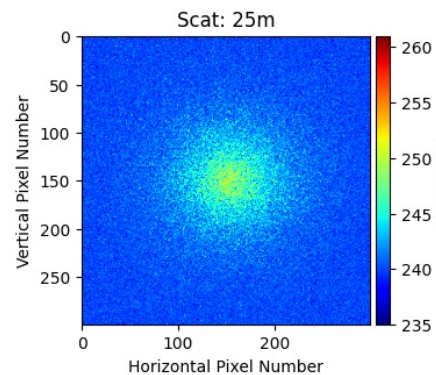
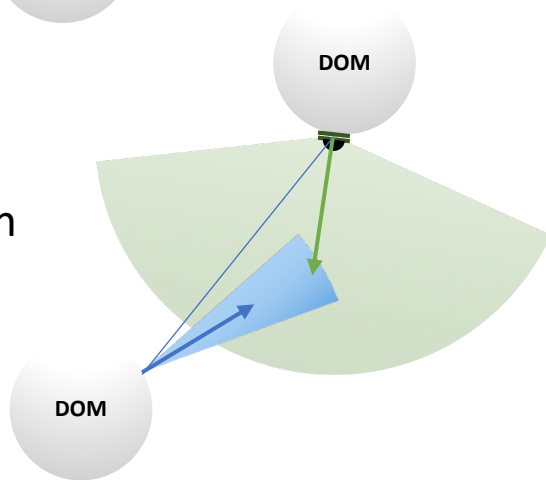
Case A

Camera and LED are
“Aligned” to each other.
In 29,8m distance

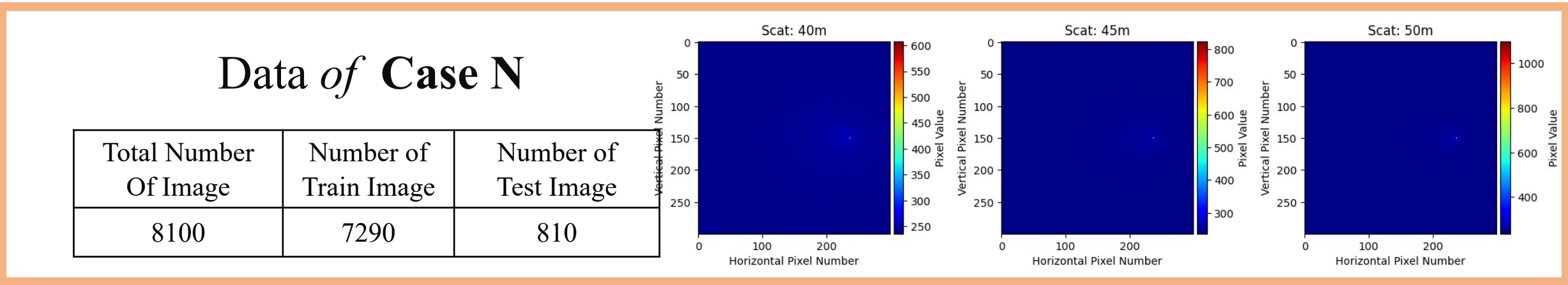
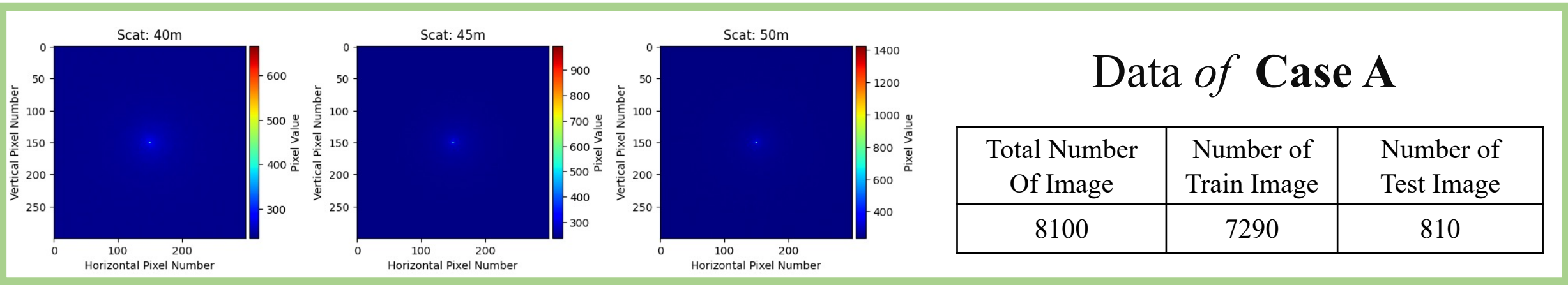


Case N

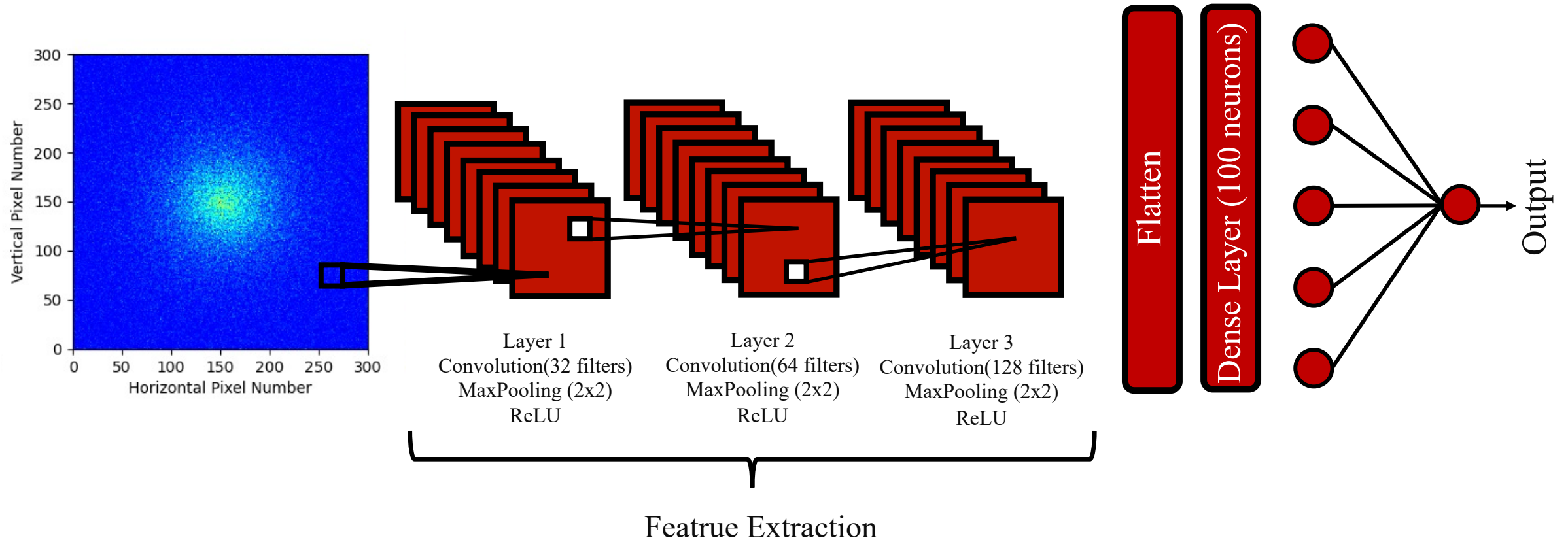
Camera and LED are
“Not aligned” to each
other with 20° 30°
orientated each.
In 29.8 distance



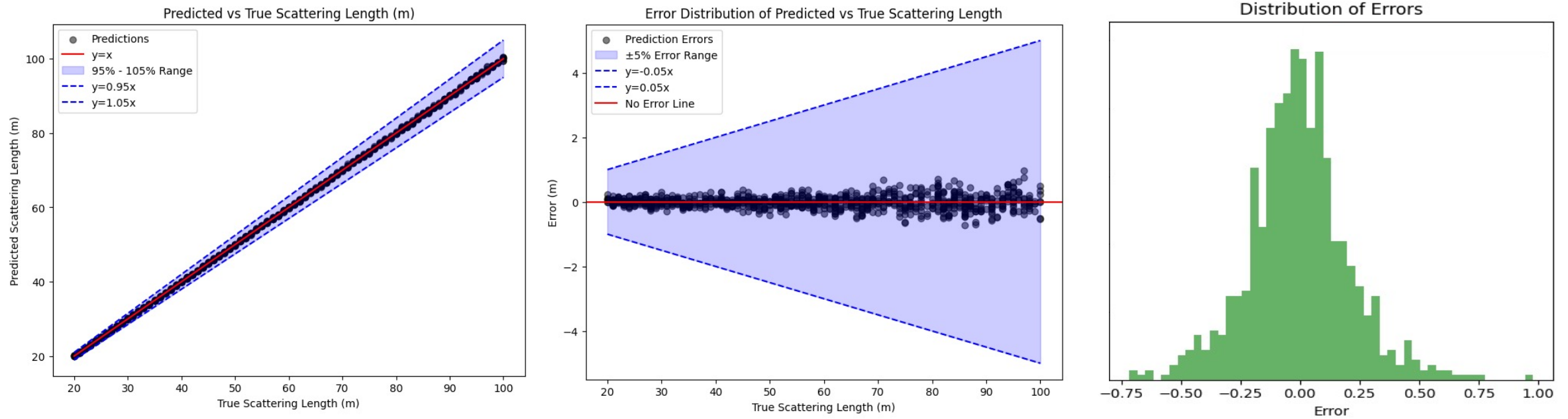
For both cases varied scattering length 20m ~100m in 1m interval
Total 81 simulations each



- Noise is added to each simulation image based on the camera's noise model.
- 100 noisy versions are generated for each simulated image. This results in a total of 8,100 (100 x 81) images.
- 810 (10% of 8,100) images are selected as test data, and the remaining images are used as train data.

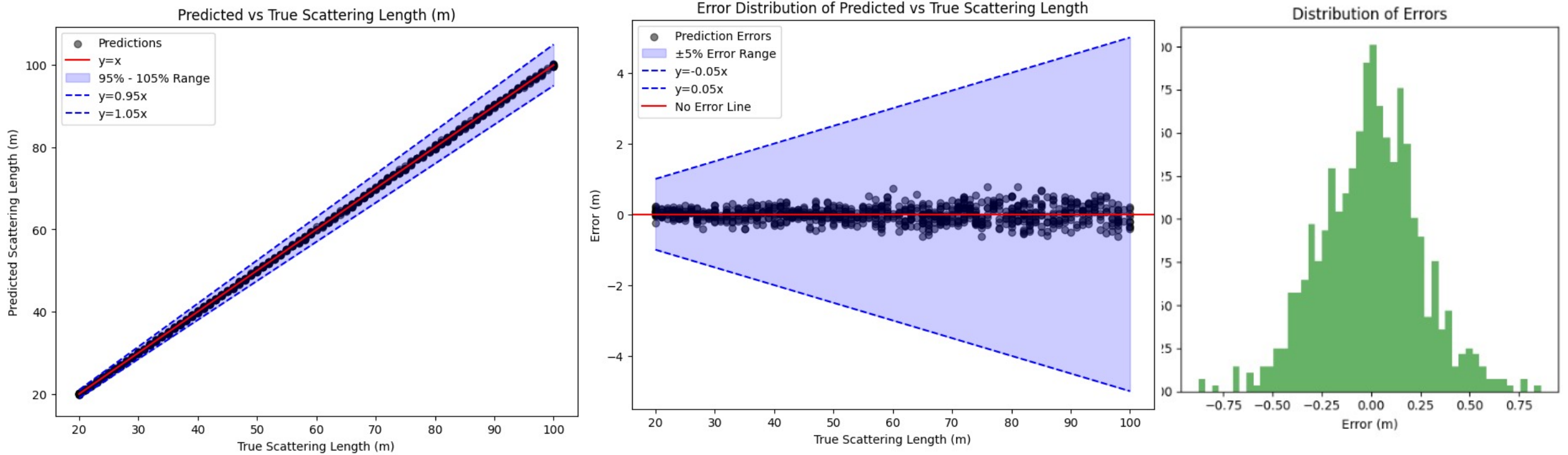


- All layers use the ReLU activation function except for the last layer, consists of 5 neurons, which uses the Leaky ReLU activation function for strengthen the nonlinearity of the network.
 - Generate Model A from Training Data of Case A
 - Generate Model N from Training Data of Case N



$$Error(m) = predicted\ scattering\ length(m) - true\ scattering\ length(m)$$

Bias(Mean Error)	Standard Deviation of error
-0.004m	0.251m



Bias(Mean Error)	Standard Deviation of error
-0.009m	0.233m

Model	Bias (Mean Error)	Standard Deviation of error
A	-0.004m	0.251m
N	-0.009m	0.233m

- The Optical properties are the largest systematic uncertainty, but the camera system has the potential to reduce this uncertainty significantly.
- Simulations of various geometric cases are underway to develop a model that can measure optical properties in any configuration.
- Developing CNN Model for Hole Ice Study measurement.

The background of the slide is a composite astronomical image. The upper portion shows a clear view of the Milky Way galaxy, with its dense band of stars and dust stretching diagonally across the dark night sky. Below the galaxy, a vibrant green aurora borealis (northern lights) is visible, creating a horizontal glow. In the foreground, the dark, silhouetted structure of a research station or observatory is visible, featuring several large cylindrical tanks and a complex network of metal scaffolding and stairs. The station is situated on a flat, dark surface, possibly a tundra or ice field.

Thank you

Feel free to ask any questions

Neutrino Astro-Particle Physics Lab

Ph.D. Candidates

Woosik Kang
Jiwoong Lee
Gyunho Yu

Master Candidates

Minyeong Seo
Seowon Choi

Under Grad Students

Minje Park
Yoonyoung Kim
Yubin Oh
Youngmin Kim



The Boss

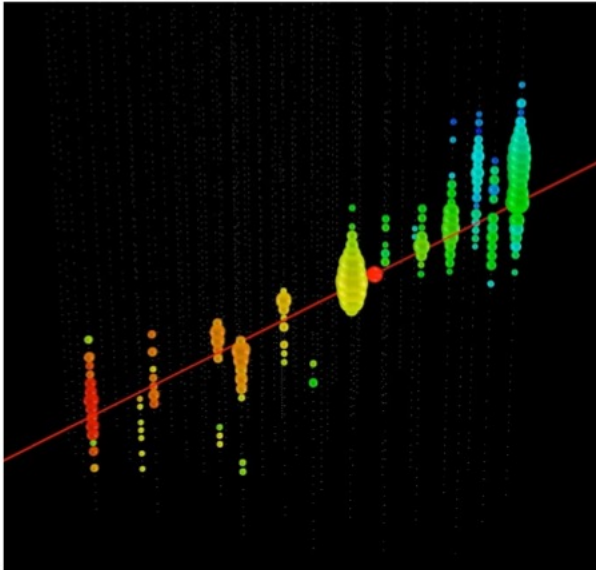
Prof. Carsten Rott

The Ph.Ds

Dr. Steven Rodan
Dr. Matti Jansson

Back Up

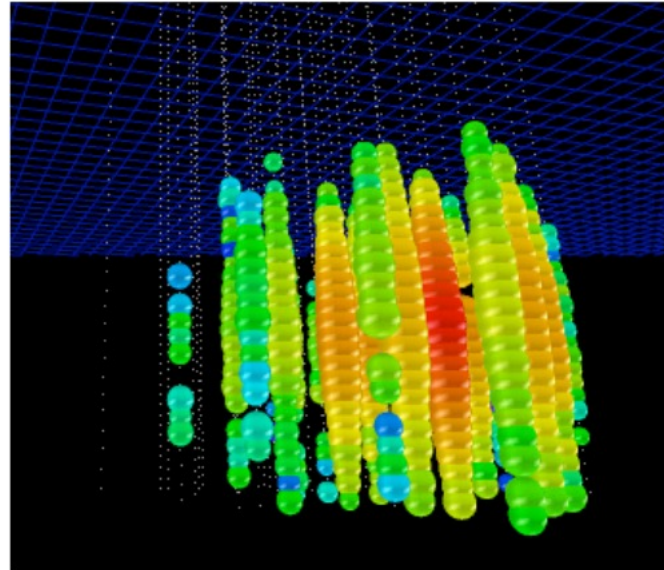
CC Muon Neutrino



Track-like

- Good direction resolution
- Bad energy resolution

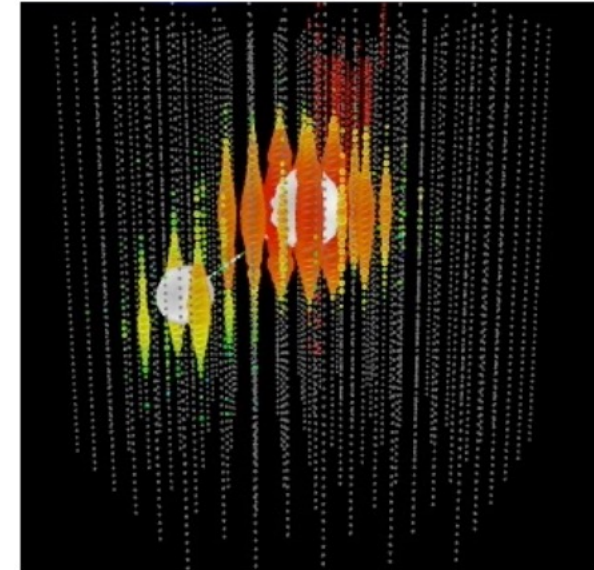
Neutral Current / Electron Neutrino



Cascade-like

- Bad direction resolution
- Good energy resolution

CC Tau Neutrino



Double Bang

- τ CC only
- (not observed yet)

Uncertainty by the Optical Properties

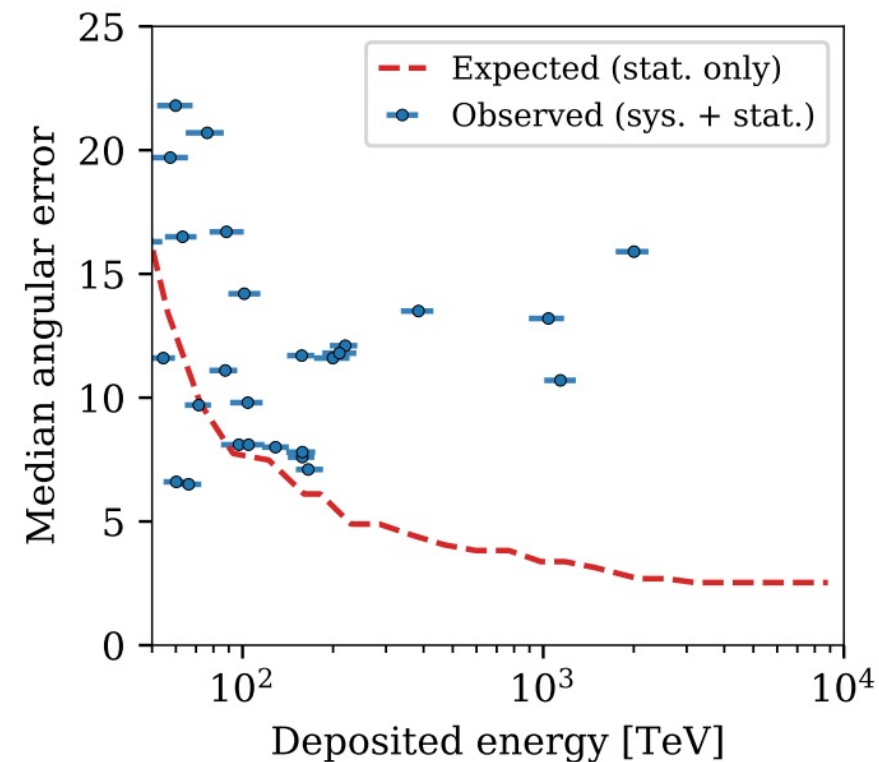
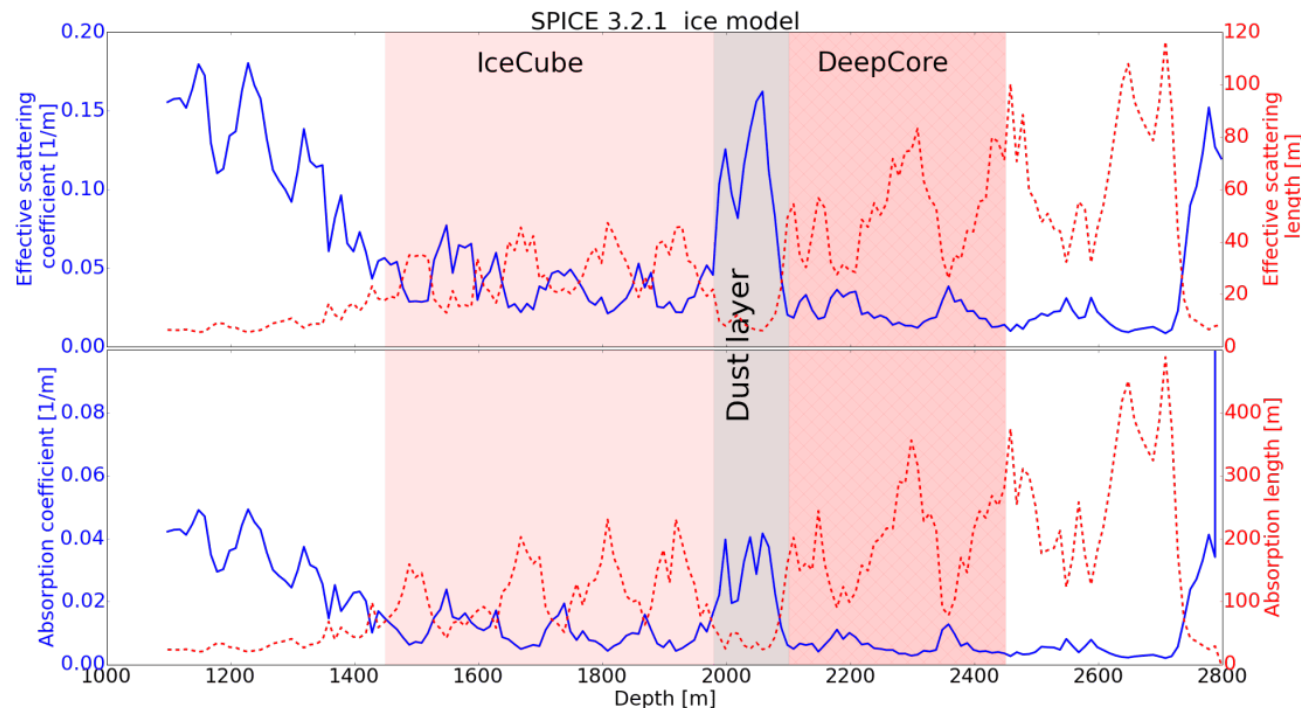
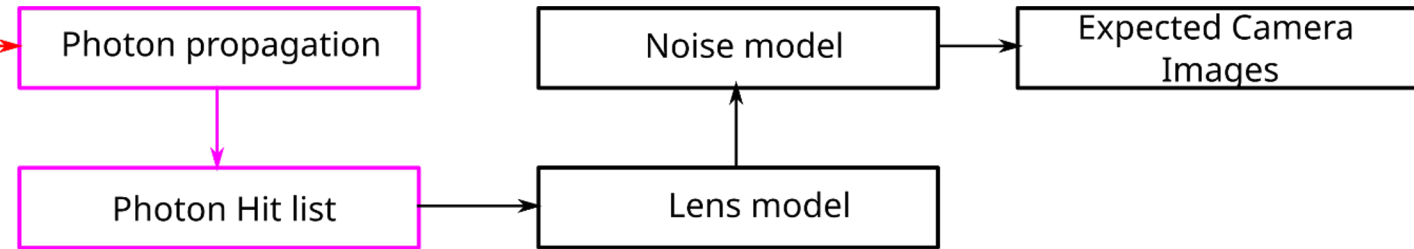
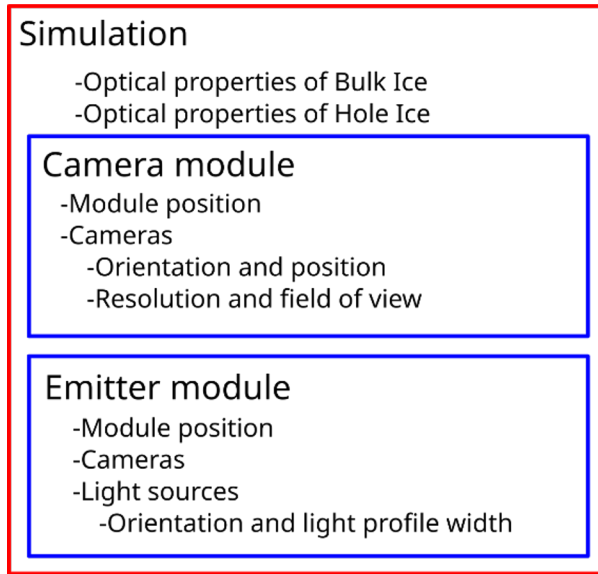
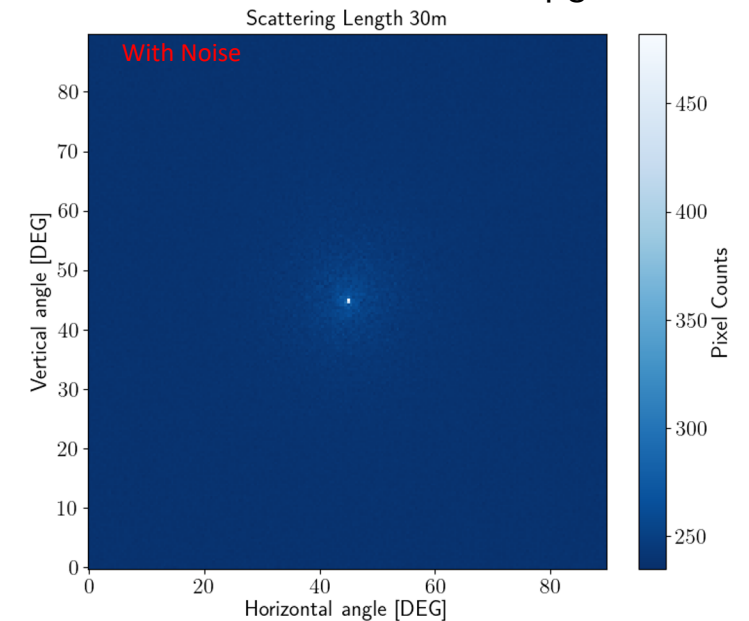
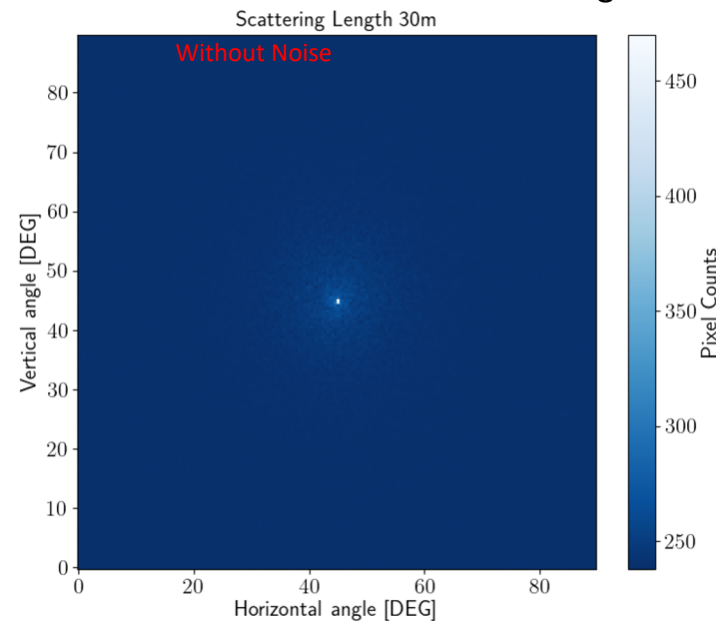
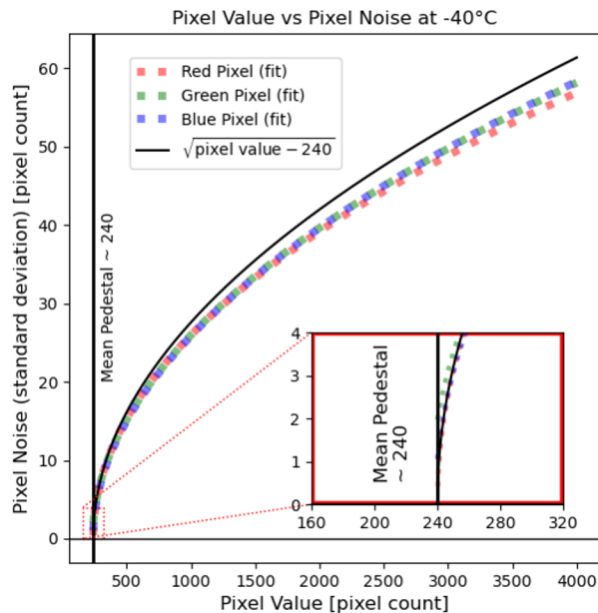


Figure 4: Observed median angular error of fully contained high energy (HESE) cascade directional reconstruction as a function of reconstructed deposited energy. The dashed line indicate the reconstruction performance with a perfect knowledge of the optical properties of ice and detector responses. The deviation of data points from the line indicate the presence of incomplete understandings of ice and detector response to bright light.



- CamSimframework uses the PPC program [1] for photon propagation with plans to use CLsim [2]
- Camsim was developed to simplify use of propagation code and generate images.
- Uses 33 cm modules with attached cameras and light sources
- The Simulation is set up with these modules and the simulated ice properties.
- It can simulate a column with different optical properties to the surrounding medium.
- Simulations of the IceCube Upgrade [3] camera and illumination system [4] have been run.
- A noise model was derived from long term measurements with cameras for the upgrade.



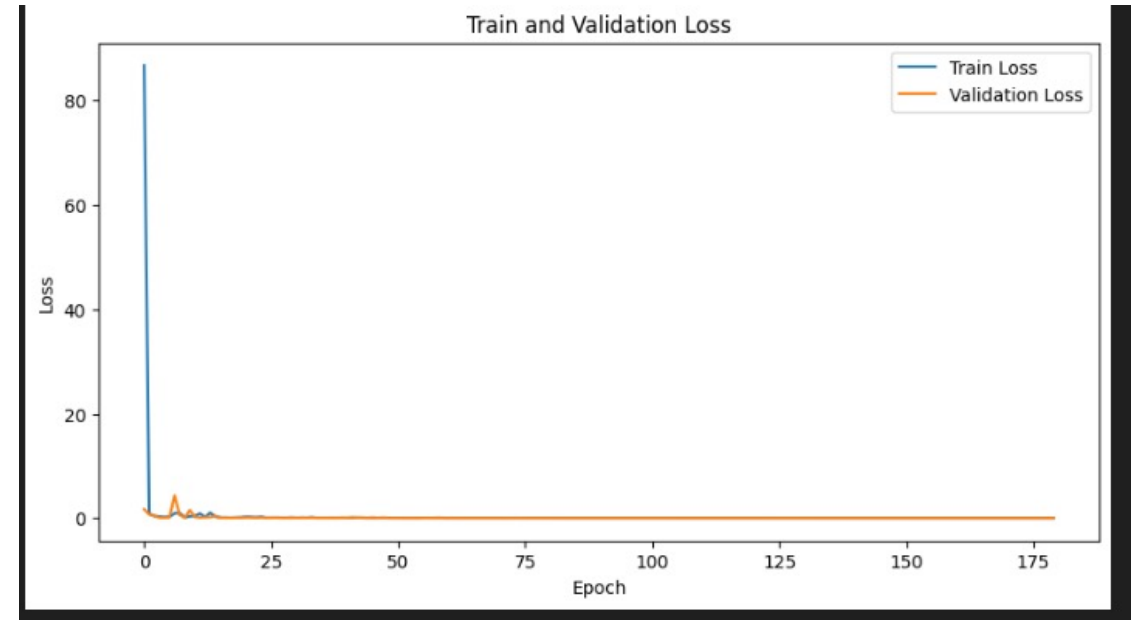
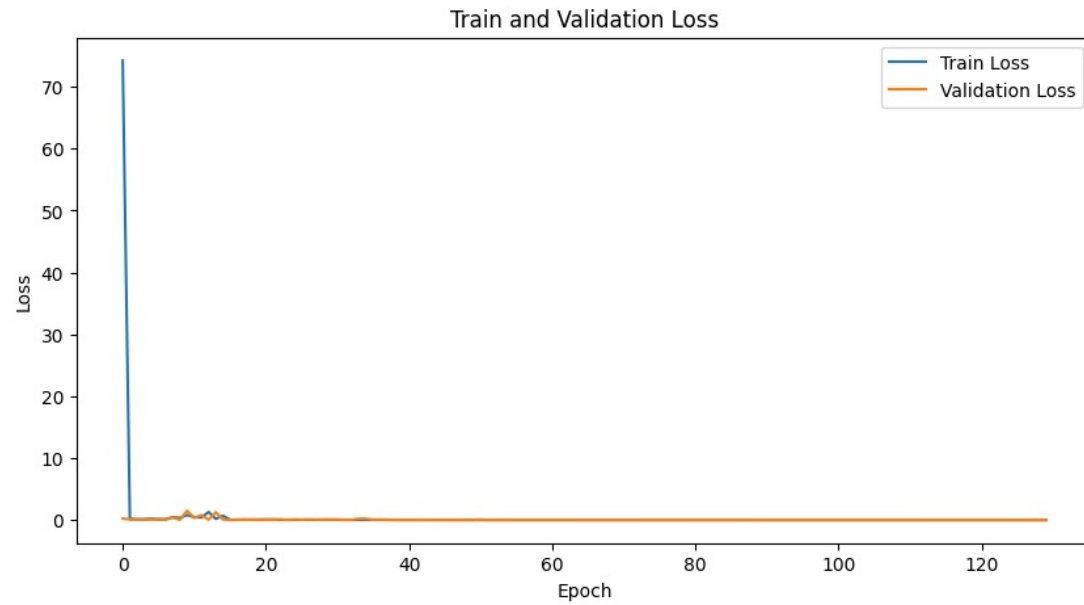
Model Summary

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Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 298, 298, 32)	320
max_pooling2d (MaxPooling2D)	(None, 149, 149, 32)	0
conv2d_1 (Conv2D)	(None, 147, 147, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 73, 73, 64)	0
conv2d_2 (Conv2D)	(None, 71, 71, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 35, 35, 128)	0
flatten (Flatten)	(None, 156800)	0
dense (Dense)	(None, 100)	15,680,100
dense_1 (Dense)	(None, 5)	505
dense_2 (Dense)	(None, 1)	6

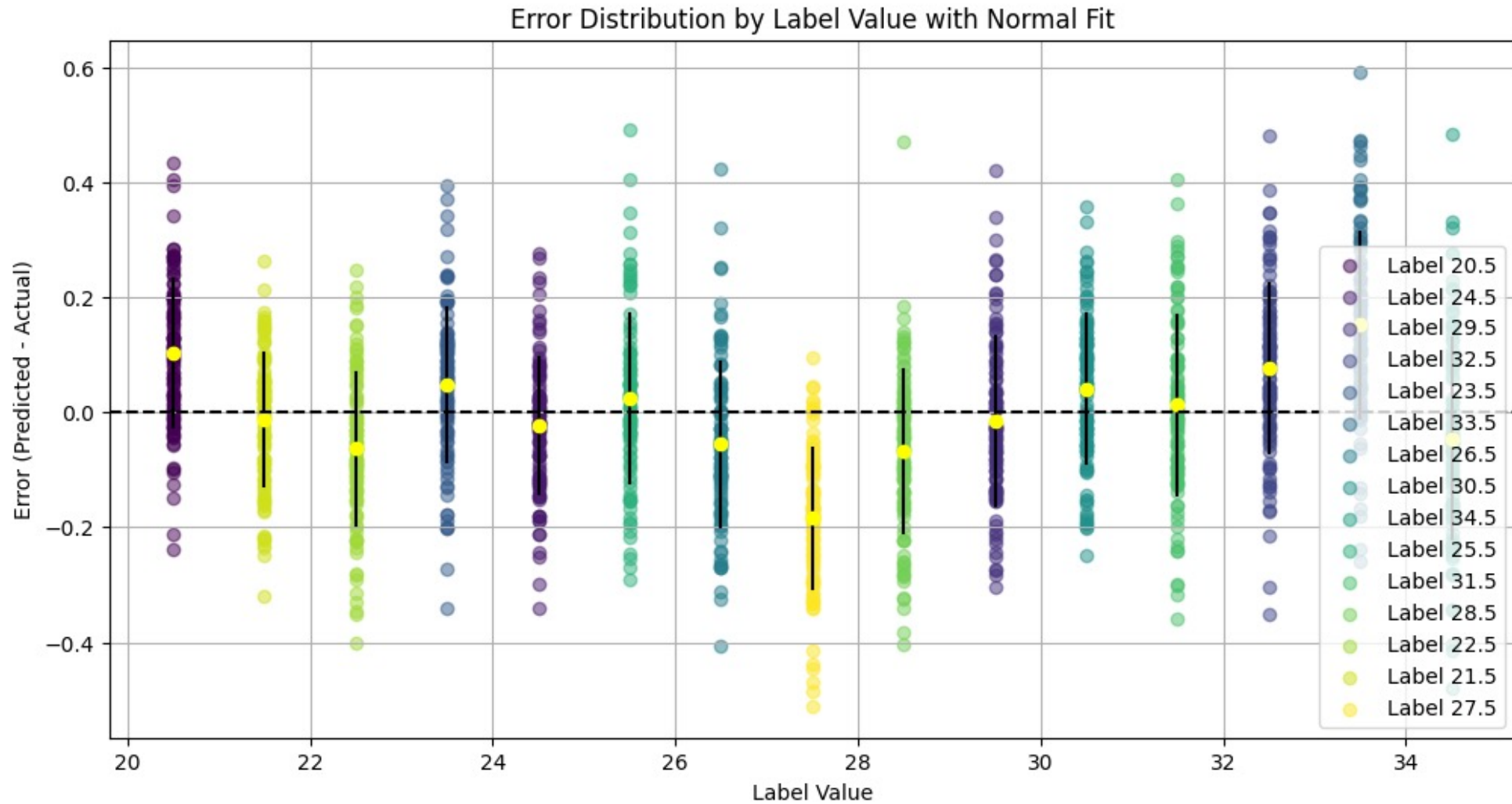
Total params: 15,773,283 (60.17 MB)

Loss function



Performance at *non-trained scattering length*

23



Performance at *non-trained scattering length*

24

```
stat, p_value = stats.shapiro(errors)
print("Shapiro-Wilk Test statistic:", stat)
print("P-value:", p_value)
```

```
# 결과 해석
if p_value > 0.05:
    print("오차 분포는 정규 분포를 따릅니다 (정규성 가정을 기각하지 않습니다).")
else:
    print("오차 분포는 정규 분포를 따르지 않습니다 (정규성 가정을 기각합니다).")
```

Shapiro-Wilk Test statistic: 0.9984514707286232

P-value: 0.19201504316096085

오차 분포는 정규 분포를 따릅니다 (정규성 가정을 기각하지 않습니다).

